Responses to Selection Criteria

Wentao Zhang · Application for Lecturer in Electrical Engineering / Biomedical Engineering · R-63679
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The responses below address each bullet of the "About You" section of the position description, mapped where relevant to the corresponding domain in UQ's *Criteria for Academic Performance* for Academic Level B (Lecturer, Teaching & Research).

1. A PhD in electrical engineering, biomedical engineering or a closely related field
I hold a **PhD in Electrical Engineering** from Southeast University (SEU), Nanjing, China (Sep 2021 – Sep 2025), with the dissertation *Multi-physics Analysis and Design of Hairpin Winding PM Traction Machines in Electric Vehicles*. My MSc (2018–2021) and BEng (2014–2018) are also in Electrical Engineering from SEU. The doctoral programme combined coursework in advanced electromagnetics, control, power electronics and finite-element methods with five years of independent research in the multiphysics design and AI-assisted analysis of electrical machines. My dissertation work was recognised by the 2025 2nd Prize Paper Award of *IEEE Transactions on Transportation Electrification* and the 2025 1st Prize in the IEEE TEC Electric Vehicle Design Competition.

2. A demonstrated record of high-quality research outputs in biomedical or electrical engineering
I have authored **22 journal papers, including 17 published IEEE Transactions** (one further in early access) across *IEEE Transactions on Industrial Electronics*, *Transportation Electrification*, *Industry Applications*, and *Energy Conversion*; twelve are first-author, including five in *Transactions on Industry Applications*, three in *Transactions on Transportation Electrification*, and two in *Transactions on Industrial Electronics*. I have a further **18 peer-reviewed conference papers** (six first-author at IEEE ECCE, IEMDC and ICEM) and hold **nine patents**. Recognition of research quality includes the 2025 2nd Prize Paper Award of *IEEE Transactions on Transportation Electrification*, the 2025 1st Prize in the IEEE TEC Electric Vehicle Design Competition, the 1st Prize in the IEEE IAS CMD Thesis Contest, the CIEEC Best Paper Award (2023), and the Outstanding Master Degree Thesis in Jiangsu Province (2022). This profile is consistent with — and at the upper end of — the Level B expectation of "research outputs consistent with discipline norms, with a lead role in some outputs, resulting in a developing national profile."

3. Demonstrated capability or potential to attract competitive research funding
My early-career profile aligns with the Level B expectation of "contributing to, and sometimes leading, funding applications." To date I have contributed to externally funded programmes as a research-team member: my doctoral research at SEU was carried out within Prof. Wei Hua's group on industry-supported electric-drive projects with major Chinese OEM and Tier-1 partners, and my current Research Assistant role at the Power Electronics and Machines Centre (PEMC), University of Nottingham, sits within an industry-aligned high-power-density motor and cooling-system programme. The competitive funding pipeline I would build at UQ has three tracks:

- **AI-for-science magnetic-field surrogate libraries** with reproducible benchmarks across electromagnetic device classes — a natural fit for ARC Discovery Projects and ARC DECRA.
- **High-fidelity multiphysics co-design frameworks** for next-generation electric drives, with industry co-funding through ARC Linkage / iLab pathways.
- **Cross-disciplinary collaborative proposals** with colleagues whose science depends on fast and accurate magnetic-field analysis, for which the operator-learning surrogate methodology I have developed is directly applicable.

I have already begun building the partner relationships needed to support these proposals, including with collaborators at the University of Sheffield (Prof. Z. Q. Zhu), the University of Bath (Prof. S. Akehurst), Tshwane University of Technology (Prof. U. B. Akuru), and Nanjing University of Aeronautics and Astronautics.

4. Evidence of effective teaching contributions at undergraduate and postgraduate levels

During my PhD at SEU, I served as a **teaching assistant and laboratory demonstrator** for undergraduate courses in electrical machines and drives — preparing tutorial materials, running laboratory sessions, marking assessments, and holding office hours. At Nottingham I provide technical guidance to MSc dissertation students on electric-machine modelling and cooling. My direct undergraduate-lecturing experience is consistent with my career stage; I bring, in addition, a strong track record of **clear technical communication** (22 journal papers and three best-paper awards from IEEE-affiliated conferences where presentation quality is assessed) and **international audience experience** through co-organising the ICEM 2026 Special Session. I am ready to teach into several of the areas listed in the position description — **introductory programming for engineering students**, **embedded and digital systems** across the undergraduate progression, and **electrical-engineering team and capstone projects** — and to grow rapidly into a coordinator role with UQ's standard Level B induction support.

5. Experience or capability in course coordination and contemporary curriculum development

I have not yet held formal course-coordination responsibility, which is consistent with my career stage and with UQ's Level B expectation that lecturers "coordinate courses." I have, however, built directly transferable curriculum assets:

- **Tutorial slides and worked-example sets** for the SEU undergraduate Electrical Machines course.
- An **open-source MCU plotting and data-acquisition toolchain in Rust** and accompanying **embedded-systems demonstrators on TI C2000 and STM32 platforms**, which can be adapted as laboratory and project material for UQ's introductory programming and embedded-systems sequence.
- **Project briefs and mentoring materials** for capstone-style electric-machine design tasks used internally by the SEU and Nottingham teams.

I am familiar with contemporary engineering-education practice: project-based and competency-based learning aligned with Engineers Australia accreditation outcomes, scaffolded software/hardware co-design, and the principled integration of generative-AI tools as both a topic of study and a learning aid. I would commit to UQ's course-coordination training in my first year and to a mentor pairing with an experienced course coordinator in the School.

6. A growing record of supervision of Honours and Higher Degree by Research students

This expectation maps to the Level B criterion of "a developing track record of achievement in supervision outcomes." During my PhD I **informally co-mentored junior PhD and MSc students** within Prof. Wei Hua's group — evidenced in the co-authorship structure of my publication record, where I led the technical formulation on multiple journal and conference papers (e.g., the WFSF, hybrid-excitation, hairpin-winding and PWM-loss-coupling sub-projects). At Nottingham PEMC I serve as a designated technical advisor to MSc dissertation students on motor cooling and surrogate modelling. I am ready and motivated to take a leading role in HDR supervision: my immediate plan at UQ would be to secure an associate-supervisor role in my first six months and apply for principal-supervisor accreditation at the earliest UQ-recognised milestone, while actively recruiting Honours and HDR candidates aligned with the research programme described above.

7. Strong communication, collaboration and stakeholder engagement skills, with proven ability to successfully lead a research team

Communication is evidenced by 22 journal papers and three best-paper awards (CIEEC 2023; IEEE-IEEC 2023; IEEE IAS CMD Thesis Contest 1st Prize) where the assessment of clarity and presentation quality is a deciding factor.

Collaboration: my co-author network spans Southeast University, the University of Sheffield (Prof. Z. Q. Zhu), the University of Bath (Prof. S. Akehurst), Tshwane University of Technology (Prof. U. B. Akuru), Nanjing University of Aeronautics and Astronautics, and the University of Nottingham. Several of these are sustained, multi-year collaborations that have produced jointly authored Transactions papers.

Stakeholder engagement and team leadership: I am **Co-Organiser and Co-Chair of a Special Session at ICEM 2026** (Covilhã, Portugal), jointly with Dr Lingyun Shao, on *Next-Generation Electrified Transport: Advances in Electric Drive Systems* — covering multiphysics optimisation, cooling and thermal manage-

ment, NVH, and power-electronics integration for automotive, eVTOL and electric-propulsion applications. The role involves scoping the programme, recruiting an international author cohort, coordinating peer review with the conference chairs, and chairing the live session. Within research projects I have repeatedly led sub-teams of co-authors through the full prototype-design / FE-validation / experimental-validation cycle, as reflected in the first-author Transactions papers in my publication list.

8. A commitment to inclusive, ethical and values-driven academic practice

My research and supervision practice align with UQ's stated values — *pursuit of excellence, integrity, mutual respect, diversity and inclusion, and creativity*:

- **Integrity**: every paper I have led uses author-order conventions agreed with co-authors before submission; my doctoral research and Nottingham work follow IEEE and institutional ethics policies for data, software and disclosure; patents list inventors in accordance with their substantive contribution.
- **Diversity and inclusion**: my collaborator network spans four continents (Asia, Europe, Africa, Australia/Oceania), and I have worked across distinctly different institutional cultures (China, the UK, and via collaborations with Australia, the EU and South Africa). I would bring this perspective to mentoring students from diverse backgrounds.
- **Equity / "relative to opportunity"**: I support the principle, embedded in UQ's *Criteria for Academic Performance*, that performance is assessed with regard to the circumstances of each individual — in my own supervision practice I would apply the same standard, recognising that early-career researchers, students with caring responsibilities, and students from under-represented backgrounds make their contributions on different timelines.
- **Australian context**: I am committed to engaging with UQ's Reconciliation Action Plan and to building teaching and research practice that recognises and responds to the perspectives of Aboriginal and Torres Strait Islander staff and students.
